

Qatar-1b

R-2648

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_140912 · created 2026-08-29T23:10:22+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2456912.87422 +/- 0.00037 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.165 +/- 0.02
Transit depth (Rp/Rs)^2	2.71 +/- 0.64 [%]
Orbital Inclination (inc)	84.52 +/- 1.31
Airmass coefficient 1 (a1)	1.015 +/- 0.018
Airmass coefficient 2 (a2)	-0.004 +/- 0.012
Scatter in the residuals of the lightcurve fit is	1.74 %
Best Comparison Star	#2 - [180, 235]
Optimal Aperture	2.89
Optimal Annulus	5.28
Transit Duration (day)	0.0719 +/- 0.0071

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.165 $\pm$ 0.02 vs 0.1463 (deviation 0.93 σ)
Duration fit vs published	0.0719 d vs 0.0692 d (deviation 0.38 σ)
Mid-time O-C	-0.8 min
Depth SNR	8.2
Residual scatter	1.74 %

Light curve

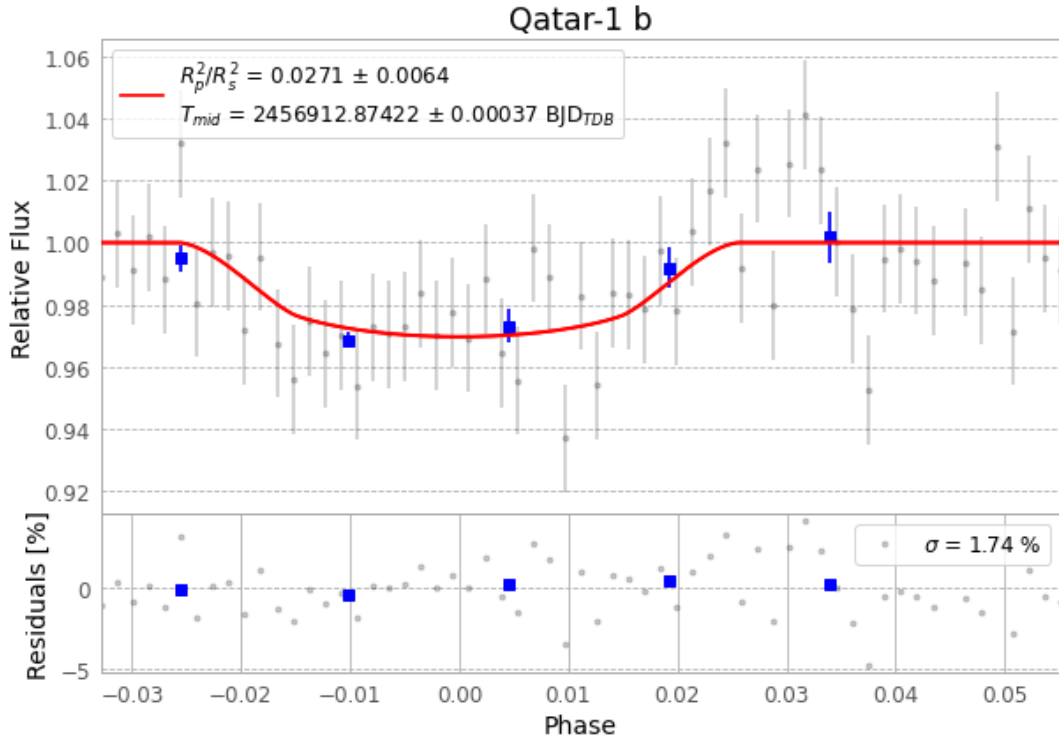


Figure 1 — FinalLightCurve_Qatar-1 b_2014-09-12.png (SHA-256 1a04b09dddef94fd...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [250, 214], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 dce947e47b1859d7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

61 FITS frames (40 MB), Qatar-1140912074813.FITS → Qatar-1140912104812.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-140912.log (9b83fee6feb6...), FinalLightCurve_Qatar-1 b_2014-09-12.png (1a04b09dddef...), FinalLightCurve_Qatar-1 b_2014-09-12.csv (e4e350a7c91b...)

Compute resources

Wall time 102.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-29 23:10Z.